LECTURE 9

Shor's Factoring Algorithm

From period finding to factoring

Example: We want to factor 2419

A key idea common to all factoring algorithms

Find $A, B \neq 0 \pmod{2419}$ s.t.

$$A \cdot B = 0 \pmod{2419}$$

How does this factor 2419 ?

$2419 = P.Q$

P divides one of A, B

Q divides one of A, B

Claim: Both P, Q do not divide either A or B.

Proof: Suppose not. Let P, Q divide A

PQ must be a factor of A

$A = 0 \pmod{PQ}$

$A = 0 \pmod{2419}$ Contradiction

g.c.d $(A, 2419)$ will be either P or Q

$$F(x) = 2^x \pmod{2419}$$

$F(0) = 1$

$F(1) = 2$

$F(2) = 4$

$F(3) = 8$

$\vdots$

$F(10) = 1024$

$F(11) = 2048$

$F(12) = 1677$

$F(13) = 935$

$\vdots$

$F(580) = 1$

$F(581) = 2$

$F(582) = 4$

$\vdots$

The period of F is 580.

How can it help to figure out factors of 2419?

$$2^{580} = 1 \pmod{2419}$$

$$(2^{290})^2 - 1^2 = 0 \pmod{2419}$$

$$(2^{290}+1)(2^{290}-1) = 0 \pmod{2419}$$

$$2^{290} = 532 \pmod{2419}$$

$$(532)^2 = 1 \pmod{2419}$$

$$533 \times 531 = 0 \pmod{2419}$$

$$\text{g.c.d.}(531, 2419) = 59$$

$$\text{g.c.d.}(\underbrace{533}_{13\times 41}, \underbrace{2419}_{59\times 41}) = 41$$

What do we need for this idea to work?

Is finding the period enough?

$2^0, 2^1, 2^2, \ldots\ldots,$

At some point $2^x = 2^y$ since there are only finitely many remainders

$2^{L/2}$ should not be -1 mod PQ

$2^{L/2}$ should be $+1 \bmod P$
and $-1 \bmod Q$

or $-1 \bmod P$
and $+1 \bmod Q$.

and L is even

For a random a, this holds w.p. $\geq \frac{1}{4}$

Exercise

Choose $a \leftarrow \{1, 2, \ldots, PQ - 1\}$

$F(x) = a^x \bmod PQ$

<u>Algorithm</u> (Factor $R = PQ$ product of two primes hardest to factor)

Let $N = 2^n > R^3$.

- Sample a randomly from $\mathbb{Z}_R$
- $F(x) := a^x \bmod R$. (output stored in m qubits)
- $|0^n\rangle \longrightarrow \boxed{H^{\otimes n}} - \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle - \boxed{Q_F^*} - \frac{1}{\sqrt{N}} \sum_{x} |x\rangle \otimes |F(x)\rangle$

$$\frac{1}{\sqrt{N}} \sum_{x} |x\rangle \otimes |F(x)\rangle \longrightarrow \boxed{\text{Measure last } m \text{ qubits}}$$

$$\frac{1}{\sqrt{A}} \left(|x\rangle + |x+L\rangle + |x+2L\rangle + \dots + |x + \left\lfloor \frac{N}{L} - 1 \right\rfloor L\rangle \right) \otimes |F(x)\rangle$$

where L = period of F

x is a random number in $\{0, 1, \dots, L-1\}$

$A = \lfloor N/L \rfloor$ or $\lceil N/L \rceil$

$$|x\rangle + |x+L\rangle + \cdots + |x + \lfloor \tfrac{N}{L} - 1 \rfloor L\rangle \xrightarrow{\boxed{DFT}}$$

$$|\chi_x\rangle + |\chi_{x+L}\rangle + \cdots + |\chi_{x + \lfloor \frac{N}{L} - 1 \rfloor L}\rangle$$

where $\chi_y(s) = \bar{\omega}^{ys}$, $\bar{\omega} = e^{-2\pi i/N}$

N is a very large integer

Amplitude on s is

$$\chi_x(s)\left(1 + \chi_L(s) + \chi_{2L}(s) + \cdots\cdots\right) \rightarrow$$

large if sL is close to a multiple of N

LEMMA
(stated without proof)

With probability > 0.25, on measurement of

$|\chi_x\rangle + |\chi_{x+L}\rangle + \ldots + |\chi_{x+\lfloor \frac{N}{L}-1 \rfloor L}\rangle$, we see

$s = \left\lfloor \frac{kN}{L} \right\rceil$ for some integer k in $\{1, 2, \ldots, L-1\}$

i.e. $\left| s - \frac{kN}{L} \right| \leq \frac{1}{2}$

$$\left| s - \frac{kN}{L} \right| \leq \frac{1}{2}$$

Recall $L < R$, $N > R^3$, i.e. $N > L^3$, $k < L$.

How do we recover L given s, N?

Can there be L, L' s.t $\left|S - \frac{kN}{L}\right| < \frac{1}{2}$ — (1)

and $\left|S - \frac{k'N}{L'}\right| < \frac{1}{2}$ — (2)

$k \in \{1, \ldots, L-1\}$, $k' \in \{1, \ldots, L'-1\}$?

$N > L^3$, $N > L'^3$

From (1) and (2)

$$\left|\frac{kN}{L} - \frac{k'N}{L'}\right| < 1 \Rightarrow \left|\frac{k}{L} - \frac{k'}{L'}\right| < \frac{1}{N}$$

$$\left| \frac{k}{L} - \frac{k'}{L'} \right| < \frac{1}{N}$$

$$\Rightarrow \left| kL' - k'L \right| < \frac{LL'}{N} < \frac{N^{1/3} \cdot N^{1/3}}{N} < 1$$

$$kL' - k'L = 0$$

$$\frac{k}{L} = \frac{k'}{L'}$$

If our algorithm finds some L^* s.t.

$$\left| \frac{k^* N}{L^*} - S \right| < 1/2$$

then L^* must be a multiple of L

How to find k, L s.t.

$$\left| \frac{s}{N} - \frac{k}{L} \right| \leq \frac{1}{2N} \quad \text{i.e.} \quad |sL - kN| \leq L/2$$

Toy examples

(1) $L \leq 50$, $N = 1{,}000{,}000$, $s = 666667$

$$\frac{k}{L} \approx 0.666667$$

$$\frac{k}{L} = \frac{2}{3}$$

(2) $s = 181818$, $N = 1{,}000{,}000$ $L \leq 50$

$$\frac{K}{L} \approx \frac{s}{N} = 0.181818$$

$$\frac{k}{L} = \frac{2}{11}$$

(3) $s = 142857$, $N = 1{,}000{,}000$, $L \leq 50$

$$\frac{K}{L} \approx \frac{s}{N} = 0.142857$$

$$\frac{k}{L} = 1/7$$

(4) $N = 1{,}000{,}000$ $\quad S = 309{,}524$, $L \le 50$

Harder to guess?

$$\frac{k}{L} \approx 0.309524$$

How can we figure out k and L?

$N = 1{,}000{,}000$, $s = 142{,}857$

Suppose we divide N by s

$$N = 7s + 1$$

$$1 = \frac{7s}{N} + \frac{1}{N}$$

$$\frac{s}{N} - \frac{1}{7} = \frac{1}{7N}$$

$N = 1,000,000$ $\quad S = 181,818$

We expect $\frac{S}{N} \approx \frac{2}{11}$ i.e. $N \approx 5.5\,S$

$$N = \underset{\uparrow\atop 909090}{5S} + \underbrace{90910}_{S_2}$$

We expect $S_2 \approx \frac{S}{2}$

$$S = 2 \cdot S_2 - 2$$

$$N = 5S + S_2 = 10S_2 - 10 + S_2 = 11S_2 - 10$$

$$\frac{S}{N} = \frac{2s_2 - 2}{11s_2 - 10} = \frac{2}{11} + \epsilon$$

$$\epsilon = \frac{22s_2 - 22 - 22s_2 + 20}{(11s_2 - 10)11}$$

$$= \frac{-2}{(11s_2 - 10)11} = -\frac{2}{11 \cdot N}$$

$$N = 1{,}000{,}000 \qquad S = 309{,}524$$

$$N = \underbrace{3 \cdot S}_{928572} + \underbrace{71428}_{S_2}$$

$$S = \underbrace{4 S_2}_{285712} + \underbrace{23812}_{S_3}$$

$$S_2 = 3 \cdot S_3 - 8$$

$$S = 4 S_2 + S_3 = 12 S_3 - 32 + S_3 = 13 S_3 - 32$$

$$N = 3s + s_2$$
$$= 39s_3 - 96 + 3s_3 - 8$$
$$= 42s_3 - 104$$

$$s = 13s_3 - 32$$

$$\frac{s}{N} \approx \frac{13}{42} = \frac{k}{L}$$

$$\left|\frac{s}{N} - \frac{13}{42}\right| < \text{small}$$

$$s_1 = \frac{13}{42}N \pm 1$$

$$N = 3s_1 + s_2$$

$$s_2 = \frac{3N}{42} \pm 3$$

$$s_1 = 4s_2 + s_3$$

$$s_3 = \frac{N}{42} \pm 13$$

$$s_2 = 3s_3 + s_4$$

$$s_4 = \pm 42$$

In general

$$s_1 = \frac{kN}{L} \pm 1$$

i.e. $s_1 L - kN = \pm L$

We get $|Ls - Nk| \leq L$

i.e. $\left| \frac{s}{N} - \frac{k}{L} \right| \leq \frac{1}{N}$

This gives us a multiple of the period for F

- Period of F is even
- $a^{L/2}$ is not -1 mod PQ

hold w.p. $\geq 1/4$ for random a

- The period finding algorithm finds s s.t. $\left| s - \frac{kN}{L} \right| < 1/2$ for some k

This happens w.p. $1/4$

Extended g.c.d. algorithm allows to find k', L' s.t. $\frac{k'}{L'} = \frac{k}{L}$

From this, we can recover L!

Generalization: Hidden Subgroup Problem

Let G be any commutative group

Let $F : G \to \{0,1\}^m$ be periodic with period H, where H is an unknown subgroup.

i.e. $\forall\, x, h \in G$ $\quad F(x) = F(x+h)$ if and only if $h \in H$.

Examples

$G = \mathbb{F}_2^n = \mathbb{Z}_2^n$

$H = \{0, L\}$

$F(x) = F(x+y)$

iff. $y \in H$

i.e. $y = 0$ or L

$G = \mathbb{Z}_N$

$H = \{0, L, 2L, \ldots, L(\frac{N}{L}-1)\}$

L divides N

$F(x) = F(x+y)$

iff $y \in H$ i.e.

$y = 0$ or L, or $2L$ or . . .

$G = \mathbb{Z}$

$H = \{0, L, 2L, \ldots\}$

$F(x) = F(x+y)$ iff $y \in H$

$F(x) = F(x+L) = \ldots$

Create $\frac{1}{\sqrt{|G|}}\sum_{x\in G}|x\rangle \longrightarrow \boxed{Q_F^*} \longrightarrow \frac{1}{\sqrt{|G|}}\sum_{x}|x\rangle\otimes|F(x)\rangle$

$\downarrow$

$\boxed{\text{Measure last m qubits}}$

$$\frac{1}{\sqrt{|H|}}\sum_{h\in H}|x+h\rangle \otimes |F(x)\rangle$$

x is a random element of G

$$\sum_{h \in H} |x+h\rangle \longrightarrow \boxed{\text{F.T. for group } G}$$

$$\downarrow$$

$$\sum_{h \in H} |\chi_{x+h}\rangle$$

Amp on s is $\chi_s(x) \sum_{h \in H} \chi_h(s)$ → learn useful info about H

One other important example : discrete log modulo a prime

given $g^x \pmod{p}$ and g,

find x.

Question : How to use H.S.P. to solve this_